How Accurate Are Learned World Models?
An Empirical Analysis of DreamerV3's Imagination on Snake

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Abstract

DreamerV3 learns a world model from experience and then trains its policy entirely inside that model—it learns by dreaming rather than by doing. This approach works across dozens of domains without changing a single hyperparameter (Hafner et al. 2025). But how accurate are those dreams? If the world model gets something wrong, how does the agent still succeed? And does imagination accuracy even matter for performance? I test this by training DreamerV3 on Snake—a game with simple, deterministic dynamics that make imagination errors easy to measure. By extracting imagined trajectories at multiple training checkpoints and comparing them against real game states, I can quantify what the world model gets right, what it gets wrong, and whether imagination accuracy actually correlates with game performance. Snake has not been studied with DreamerV3 before, and no prior work has systematically measured imagination fidelity in this way.

Do not use first person singular pronoun in formal technical writing.

Introduction

Model-based reinforcement learning takes a fundamentally different approach from model-free methods. Instead of learning a policy directly from environment interactions, model-based agents first learn how the environment works and then use that learned model to plan (Sutton and Barto 2018). The Dreamer family of algorithms takes this idea further—the agent trains its policy entirely inside the learned world model, never touching the real environment during policy optimization (Hafner et al. 2020, 2021, 2025).

DreamerV3 represents the current state of the art in this approach. It masters Atari games, continuous control tasks, and even collects diamonds in Minecraft, all with the same fixed hyperparameters (Hafner et al. 2025). However, most research on DreamerV3 focuses on end-task performance—how high the score is, how fast the agent learns. What gets less attention is the world model itself. The agent is training inside an imagined version of the environment. So how faithful is that imagination to reality?

This is not just an academic question. If the world model predicts the wrong outcome for an action, the agent is optimizing for a scenario that does not exist. Understanding when and how imagination diverges from reality could reveal fundamental insights about why model-based RL works as well as it does—and where it breaks down.

The central question is: **Does DreamerV3 need accurate imagination to learn effective policies, or can it succeed even when its world model is wrong?** I break this into three testable hypotheses:

H1: The world model's imagination gets more accurate as training goes on—it learns to dream better over time.

H2: Better imagination leads to better play. If the dreams are more accurate, the agent scores higher.

H3: Getting the game state right matters more than getting the pixels right. Knowing where the snake and food are is more important than producing a sharp image.

If H2 holds, imagination quality is a bottleneck—improve the world model and the policy improves with it. But if H2 is weak or flat-out wrong, that means the world model is capturing the right structure for making decisions without actually being a faithful picture of reality. That would go a long way toward explaining why DreamerV3 works across so many different domains without changing anything.

I propose to investigate this by training DreamerV3 on Snake. Snake is an ideal testbed for imagination analysis because its dynamics are simple and deterministic: the snake moves one cell per step in the current direction, food appears at random locations, and the game ends on collision with a wall or the snake's own body. This simplicity means I can precisely identify what the world model gets right and what it gets wrong. There is no visual clutter, no stochastic physics, no ambiguity. When the model imagines a trajectory, I can compare it frame by frame against what would actually happen.

The contributions of this work are:

1. A systematic analysis of DreamerV3's imagination accuracy on Snake, measuring how well imagined trajectories match real game states across training
2. Evidence on whether imagination accuracy correlates with task performance—does the agent need accurate dreams to play well?
3. An element-level breakdown of what the world model learns to predict (snake position, food location, collisions) and what it struggles with.

It will be good to try to make your contributions more general.

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Background

World Models in Reinforcement Learning

The idea of learning a model of the environment and using it for planning has a long history in RL (Sutton and Barto 2018). Ha and Schmidhuber (2018) demonstrated that an agent could learn a compressed representation of its environment using a variational autoencoder and a recurrent network, then train a policy entirely within that learned model. This showed that "dreaming" about an environment could be enough to learn effective behavior.

Hafner et al. (2019) formalized this with the Recurrent State-Space Model (RSSM), which maintains both a deterministic recurrent state and a stochastic latent state. The deterministic path captures persistent information across time steps, while the stochastic path models uncertainty about the environment. This architecture became the backbone of all subsequent Dreamer agents.

The Dreamer Family

DreamerV1 introduced the idea of training actor-critic policies entirely within imagined trajectories generated by the RSSM (Hafner et al. 2020). DreamerV2 improved this with discrete latent representations, replacing the Gaussian latents with categorical variables, which produced sharper world model predictions and achieved human-level performance on the Atari benchmark (Hafner et al. 2021).

DreamerV3 further refined the approach with symlog predictions for handling varying reward scales and a fixed set of hyperparameters that work across diverse domains (Hafner et al. 2025). It was the first single algorithm to collect diamonds in Minecraft from scratch—a task requiring long-horizon planning across a complex open world. The key to DreamerV3's generality is that it treats world modeling as a separate problem from policy optimization, letting the same architecture adapt to radically different environments.

Related Work on World Model Analysis

Several works have extended or modified the Dreamer architecture. Deng, Jang, and Ahn (2022) removed the reconstruction objective entirely, training the world model with prototypical representations instead. Yuan et al. (2023) extended the RSSM to handle irregularly sampled data using ordinary differential equations. Sekar et al. (2020) used the world model for self-supervised exploration without task rewards. Lin et al. (2024) incorporated language into the world model, enabling agents to use instructions and descriptions for planning.

However, these works focus on improving the world model or the policy—not on measuring how accurate the imagination actually is. Walker et al. (2023) investigated model-based learning's role in exploration and transfer, and Sullivan et al. (2023) analyzed whether DreamerV3's design choices transfer to model-free methods. But a direct, systematic measurement of imagination fidelity—comparing dreamed trajectories against ground truth—remains unexplored.

Why is this important?

Snake as an MDP

I formalize Snake as a Markov Decision Process $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$:

State Space $\mathcal{S}$: Each state is a grid of size $N \times N$ containing the snake body segments, the head position, the current direction, and the food location. States are rendered as RGB images for DreamerV3's visual encoder.

Action Space $\mathcal{A}$: Four discrete actions—up, down, left, right. The snake cannot reverse direction (moving left while heading right has no effect).

Transition Function $P(s'|s, a)$: Transitions are deterministic given the current state and action. The snake moves one cell in the chosen direction. If it eats food, it grows by one segment and new food spawns at a random empty cell. The only stochasticity is food placement.

Reward Function $R(s, a, s')$: $+1$ for eating food, -1 for collision (game over), 0 otherwise.

Discount Factor γ: Set to 0.999 to encourage long-horizon survival and food collection.

The simplicity of these dynamics is precisely why Snake is a good choice for imagination analysis. Every transition except food spawning is fully predictable, so I can measure exactly where the world model's predictions diverge from reality.

Proposed Methods

Training DreamerV3 on Snake

I will implement a Gymnasium-compatible Snake environment and train DreamerV3 using the publicly available codebase. The environment will render the game as a small RGB image (e.g., 64×64 pixels) with distinct colors for the snake head, body, food, and background. I will train on a standard grid size (e.g., 10×10) and collect performance data over the full training run.

Extracting Imagined Trajectories

At regular checkpoints during training (e.g., every 50,000 environment steps), I will:

1. Record a real trajectory by running the current policy in the actual environment
2. Feed the initial observation from that trajectory into the world model
3. Roll out the world model using the same sequence of actions, generating an imagined trajectory
4. Compare the two trajectories frame by frame

This gives us matched pairs of real and imagined game states at each checkpoint, using identical action sequences so the only difference is the world model's prediction accuracy.

Measuring Imagination Accuracy

I will evaluate accuracy at two levels:

Frame-level metrics: Mean Squared Error (MSE) and Structural Similarity Index (SSIM) between imagined and real frames. These capture overall visual fidelity of the world model's predictions.

Semantic metrics: Since Snake's state is fully interpretable, I can extract game-specific information from both real and imagined frames:

- Snake head position error (Euclidean distance)
- Snake body segment accuracy (percentage of correctly placed segments)
- Food location accuracy (correct or not)
- Collision prediction accuracy (does the model correctly predict game-over states?)

These semantic metrics go beyond pixel similarity. The world model might produce slightly blurry images (low SSIM) while still correctly representing the snake's position and the food location. Conversely, it might produce sharp images that place the snake in the wrong spot. Separating these tells me what the model actually learns.

Correlation Analysis

I will correlate imagination accuracy with game performance across checkpoints. If the agent's score improves while imagination accuracy stays flat, that suggests the policy can succeed despite an imperfect world model. If accuracy and performance track together, that suggests accurate imagination is critical. I will also look at whether specific semantic metrics (e.g., food location accuracy) are more predictive of performance than others.

Evaluation Plan

Metric	What It Measures
Game score	Average food collected per episode
Survival length	Average steps before collision
MSE	Pixel-level imagination error
SSIM	Structural imagination fidelity
Head position error	Spatial accuracy of predictions
Body accuracy	Percentage of correct segments
Food accuracy	Whether food location is correct
Collision accuracy	Game-over prediction correctness

Table 1: Evaluation metrics. Frame-level metrics (MSE, SSIM) measure overall visual quality. Semantic metrics measure whether the model captures the game state correctly regardless of visual quality.

I will evaluate across 1,000 episodes at each checkpoint and report means with confidence intervals. Checkpoints will be taken every 50,000 steps over a total training run of 1 million environment steps, giving 20 snapshots of the world model's evolution.

Timeline

Date	Milestone
Feb 17–19	Proposal presentation
Feb 19	Proposal paper due
Mar 1	Snake env, DreamerV3 training
Mar 15	Imagination extraction pipeline
Mar 31	Checkpoint analysis complete
Apr 1	Midterm presentation/paper
Apr 15	Correlation & semantic metrics
May 7	Final presentation
May 12	Final paper due

Expected Challenges

Environment integration: DreamerV3 expects Gymnasium-compatible environments with image observations. Building a Snake environment that renders clean, consistent frames at the right resolution will require careful implementation.

Trajectory alignment: Imagined trajectories may diverge from real ones after just a few steps, especially early in training when the world model is poor. I will need to handle cases where the imagined game ends at a different time step than the real one.

Semantic extraction: Pulling game-state information (snake position, food location) from rendered frames requires either access to the underlying game state during real play or a reliable extraction method from pixels. I plan to log the true game state alongside rendered frames to avoid this issue.

Computational cost: DreamerV3 training is GPU-intensive. I have access to an RTX 5070 Ti with 16GB VRAM and PyTorch 2.7.0 with CUDA 12.8, which should be sufficient for a 64×64 observation space.

Conclusion

DreamerV3 succeeds by learning to imagine its environment and training inside that imagination. But how accurate those imaginations are—and whether accuracy even matters for performance—remains an open question. By applying DreamerV3 to Snake, a game simple enough to make imagination errors precisely measurable, I can provide the first systematic analysis of world model fidelity in the Dreamer framework. This work will reveal what DreamerV3's world model actually learns, how its imagination evolves during training, and whether better dreams lead to better play.

AI Usage Statement

Claude (Anthropic) was used to assist with formatting ideas into prose and structuring this document in AAAI conference format. All research questions, methods, and analysis decisions are my own.

the authors

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Formatting: As requested

Paper components: All there, well-structured

- Writing quality: Good; remove all occurrences of first person singular pronoun.

Problem: Interesting, good research questions/hypotheses

Abstract: Good, appropriate

Intro + Motivation: Good, I will like the writing/presentation to be more general, if possible

Related Work: From very good sources, good quantity

Datasets: Domain identified, dataset will be created

Approach: Outlined well, generic

Evaluation metrics: Identified, appropriate

Timeline: Given

Conclusion: There

References: High quality sources, recent

A- 2/20/26

You can remove all 'I's, 'my's, etc., and resubmit within a week for better grade.